

SUPPLY CHAIN DELIVERY PERFORMANCE ANALYSIS

End-to-End Business Report & Strategic Recommendations

Report scope	Delivery performance, profitability impact, operational bottlenecks and predictive risk
Dataset	DataCo Smart Supply Chain Dataset (Kaggle) — 180,519 raw orders × 53 columns
Analysis period	January 2015 – February 2018
Orders analysed	176,827 orders after removing 3,692 cancelled orders and 33 unused columns
Tools	Python 3 — pandas, NumPy, Matplotlib, Seaborn, scikit-learn, imbalanced-learn (SMOTE)
Companion artefact	Supply_Chain_Delivery_Performance_Analysis.ipynb (executed notebook, 6 charts)
Date	27 September 2026

54.71%	\$2.1M	100%	73.7%
of orders arrive after the promised window	profit at risk on delayed, loss-making orders	delay rate of First Class shipping	predictive model accuracy on unseen orders

Bottom line. More than half of all orders (54.71%) miss their promised delivery window. Delays put **\$2.1 million** of profit at risk, one in five orders (18.7%) is loss-making, and the root causes are fully identifiable and addressable: shipping-mode misconfiguration (First Class = 100% delay), payment-processing friction, and seasonal peaks that capacity planning does not cover. A Random Forest model already flags at-risk orders with 74% accuracy — the fix is operational, not aspirational.

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1. Executive Summary

This report presents an end-to-end analysis of delivery performance for a global e-commerce operation selling sporting goods, fitness equipment, outdoor gear, footwear and apparel across multiple international regions. The analysis covers **176,827 orders** placed between **January 2015 and February 2018**, cleaned and engineered from a 53-column master dataset down to 20 analytical features.

54.71% of orders (96,737 shipments) fail to meet their promised delivery window. Late delivery is therefore not an edge case — it is the default experience for the majority of customers. Delayed, loss-making orders collectively carry **\$2.1 million of profit at risk**, and even the worst decile of delays is contained within **3 days**, confirming a systematic, process-driven problem rather than a catastrophic one.

Order-level profitability splits into **80.6% profitable, 18.7% loss-making and 0.6% breakeven**. Bottleneck detection across six operational dimensions isolates **First Class shipping (100% delay rate)** as the single largest structural failure, while every region, customer segment and payment status clusters tightly between 53% and 59% delay — evidence that the issue is company-wide, not local.

A Random Forest classifier trained on order-time features with SMOTE-balanced training data achieves **73.7% accuracy, 77.9% precision and 74.1% recall** on unseen orders, giving the business a deployable early-warning system that identifies at-risk orders the moment they are placed.

Key findings at a glance

- **Majority failure:** 54.71% late vs 45.29% on time — a structural, not exceptional, outcome.
- **Financial drag:** \$2.1M of profit sits on delayed, loss-making orders; average order profit is only \$22.01.
- **Shipping mode is the #1 driver:** First Class 100% late, Second Class 79.8% late, Standard Class 39.7%, Same Day 0%.
- **Payment friction:** PAYMENT_REVIEW, PENDING and PENDING_PAYMENT statuses run 15–25 points above baseline in the worst region.
- **Seasonality:** delays peak in August (55.4%), September (55.3%) and December (55.4%); July is the healthiest month (53.9%).
- **Time of order:** evening orders at 20:00 reach 57.4% delay, with a secondary 12:00–14:00 bulge; Mondays (55.6%) and Sundays (55.1%) are worst.
- **Profit erosion:** orders shipped ahead of schedule generate the highest total and mean profit — lateness destroys value order by order.
- **Predictable risk:** a 74%-accurate Random Forest model can flag at-risk orders before dispatch.

Recommended action set (summary): audit First/Second Class routing rules, deploy the predictive alert at order capture, automate escalation of payment reviews beyond two hours, build seasonal capacity for August–December, default eligible orders to Standard Class, and retrain the model quarterly targeting >80% recall on late orders.

2. Business Context & Analytical Objective

A supply chain is the end-to-end network that follows a product from raw materials, through manufacturing, warehousing (inventory) and distribution, to the final customer. Every step generates data — cost, demand, inventory and delivery-time analysis are the four classic forms this work takes. **Delivery time is the most sensitive of them:** it is directly proportional to customer satisfaction, because a promise made at checkout and broken in transit erodes trust faster than almost anything else.

Business problem

Actual shipping times frequently deviate from the scheduled delivery windows, creating eroded customer trust, reduced order profitability, and an inability to make reliable commitments to the buyer at the point of purchase. The company needs a root-cause diagnosis of *why* deliveries slip, a quantification of *how much it costs*, and a set of actionable recommendations to close the gap.

Analytical objectives

- Understand the **current state of delivery performance** across all operational dimensions.
- **Quantify the financial impact** of delays on order profitability.
- Identify the **primary operational bottlenecks** driving late deliveries.
- **Build a predictive model** that flags at-risk orders before they become late.
- **Deliver actionable recommendations** to improve on-time delivery and profitability.

Data preparation summary

Step	Action	Result
Load	DataCo master dataset, latin-1 encoding	180,519 rows × 53 columns
Quality check	Duplicate scan; missing-value scan	0 duplicates; 4 columns with nulls
Clean	Drop 33 unused/PII/ID/redundant columns (Product Description, e-mail, passwords, lat/long, IDs, Benefit per order...)	20 analytical columns
Filter	Remove 3,692 CANCELED orders (no delivery performance to measure)	176,827 orders
Types	Convert the two DateOrders columns to datetime	0 missing values
Engineer	Order Processing Time, Delay, Is Delayed, Order Month / Day / Hour, Profitability	5 analytical features added

3. KPI Scorecard

These are the eight indicators reviewed with stakeholders before any dashboard is built — the Python equivalent of the KPI cards that later appear in Power BI / Tableau.

176,827	96,737	\$7.69M	\$2.11M
total orders analysed	late deliveries (54.71%)	total profit from profitable orders	profit at risk on delayed orders

Business KPI	Value	Interpretation
Total orders	176,827	Clean analytical base, 2015–2018
Late deliveries	96,737	Orders dispatched beyond the scheduled window
Late delivery rate	54.71%	Majority of customers experience a late order
On-time delivery rate	45.29%	Fewer than half of orders meet the promise
Total profit (orders > 0)	\$7.69M	Gross positive contribution of profitable orders
Profit at risk (delayed)	–\$2.11M	Negative profit concentrated on delayed orders

Business KPI	Value	Interpretation
90% of delays within	3 days	Problem is systematic, not catastrophic
Average order profit	\$22.01	Thin margin — every avoidable failure matters

What this means: the operation is losing money on nearly one order in five while failing its delivery promise to more than one customer in two. Because the 90th percentile of delay is only 3 days, the levers are process configuration and scheduling — not network redesign.

4. Profitability Analysis

Order-level profit was classified into three tiers — **Profit** (> 0), **Loss** (< 0) and **Breakeven** ($= 0$) — to establish where value is created and destroyed across the order book.

Profitability Distribution of Orders

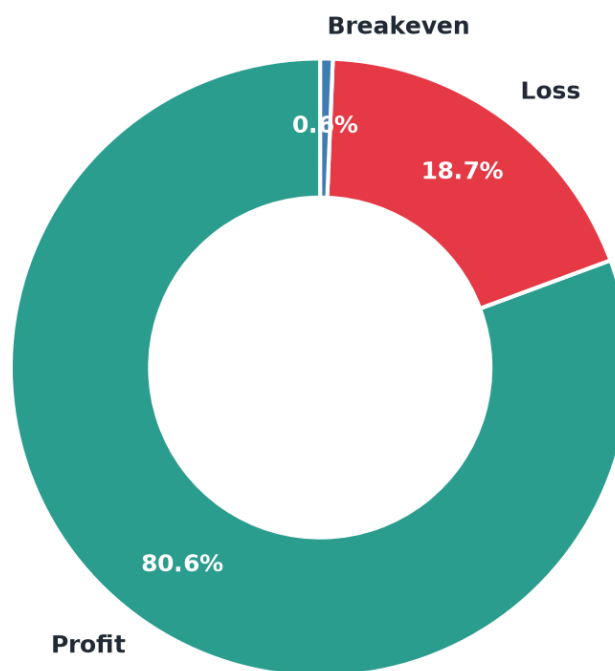


Figure 1 — Profitability distribution of orders (n = 176,827)

Tier	Orders	Share
Profit	142,590	80.6%
Loss	33,088	18.7%
Breakeven	1,149	0.6%

Roughly **four out of five orders are profitable**, but the residual **18.7% loss-making share** is large for a business whose average order earns only \$22.01. Every loss-making order is effectively subsidised by two profitable ones, which makes the loss tail — heavily populated by delayed orders — the natural target for intervention.

5. Profitability vs Delivery Delay

Orders were grouped by delay bucket (days beyond the scheduled window) to test whether lateness actually costs money. The delay field also captures early shipments (–1 and –2 days), which serve as the natural baseline.

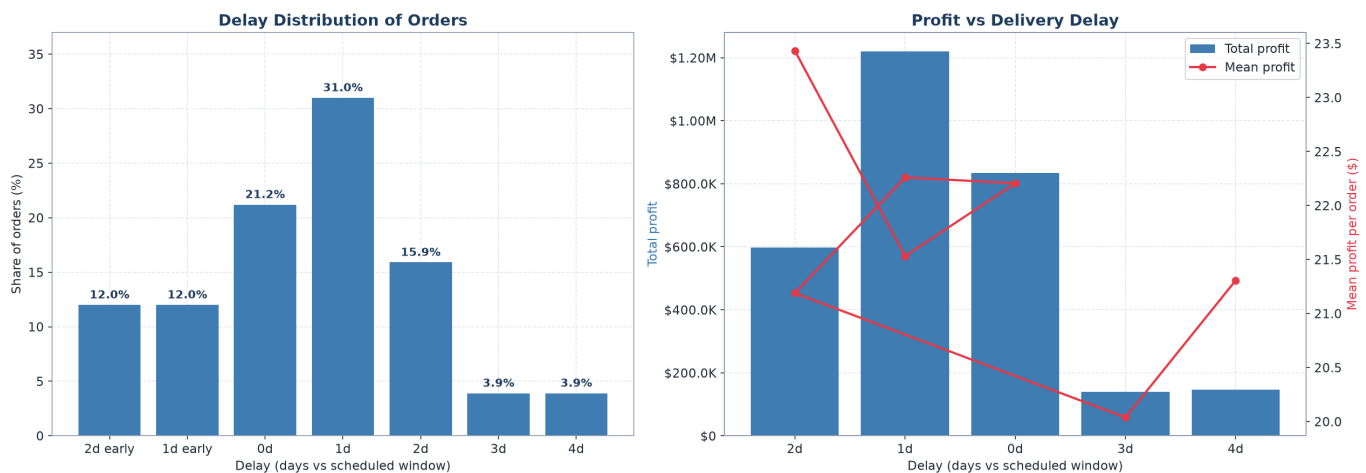


Figure 2 — Left: distribution of orders by delay. Right: total profit (bars) and mean profit per order (line) by delay bucket.

- **Delay distribution:** 21.2% of orders ship exactly on schedule, 31.0% are one day late, 15.9% two days late, and 3.9% + 3.9% three and four days late; a further 24% ship 1–2 days early.
- **Early = profitable:** orders shipped ahead of schedule carry the highest total profit, and mean profit per order declines steadily as the delay grows (from ~\$23 to ~\$20 at three days late).
- **Abnormal tail:** the four-day bucket shows a small mean-profit uptick — a handful of high-value orders, not a genuine pattern.
- **Financial reading:** mean profit per order stays in a narrow \$20–\$23 band, so the damage is driven by *volume* of late orders rather than a single catastrophic cohort — recovering even a few points of on-time rate converts directly into margin.

6. Bottleneck Detection

Delay percentage was computed across six operational dimensions to locate where problems actually occur. Each dimension is reduced to a single comparable metric: *late orders ÷ total orders*.

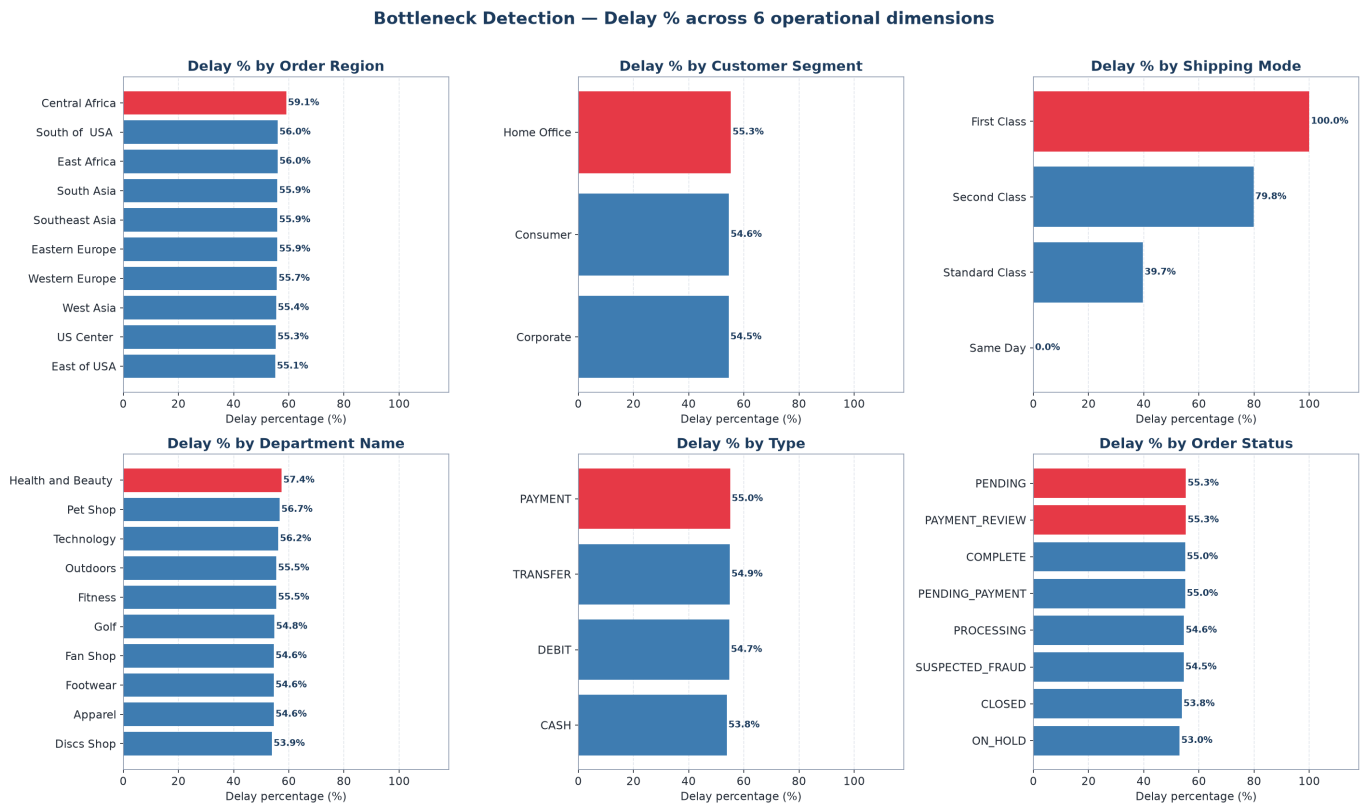


Figure 3 — Delay percentage across six operational dimensions (worst level highlighted in red).

Dimension	Worst-performing level	Delay %	Reading
Shipping mode	First Class	100.0%	Structural misconfiguration — every First Class order is late
Shipping mode	Second Class	79.8%	Express tiers both fail; Standard Class is only 39.7%
Order region	Central Africa	59.1%	Highest region, but all regions cluster at 53–59%
Customer segment	Home Office	55.3%	No segment receives preferential service
Department	Health and Beauty	57.4%	Pet Shop 56.7%, Technology 56.2% follow closely
Payment type	PAYMENT	55.0%	All four payment types sit within 1.2 points of each other
Order status	PENDING / PAYMENT_REVIEW	55.3%	Payment states sit at the top of the status list

Two conclusions. (1) Because regions, segments, statuses and payment types all cluster between 53% and 59%, the delay problem is **systematic and company-wide** — it rules out localised logistics failures. (2) **Shipping mode is the genuine outlier:** First Class at 100% and Second Class at 79.8% vs Standard Class at 39.7% and Same Day at 0% prove that express service levels are being promised with a scheduling configuration the network cannot honour.

7. Root Cause Analysis — Top Drivers

Bottleneck detection told us *where* the problem is worst; root-cause analysis drills into the worst region (**Central Africa, 59.1%**) and ranks every factor level — shipping mode, customer segment, department, payment type and

order status — by delay percentage.

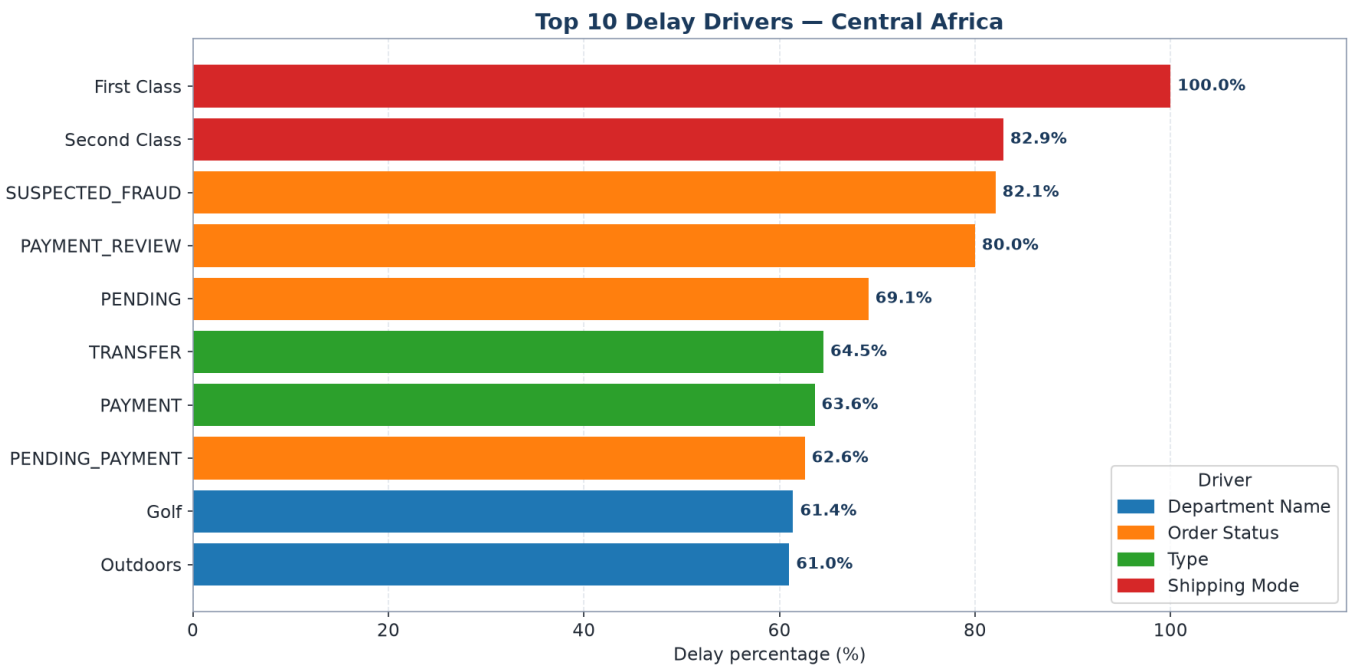


Figure 4 — Top 10 delay drivers inside Central Africa.

Central Africa — top 10 drivers

#	Driver	Factor level	Delay %
1	Shipping mode	First Class	100.0%
2	Shipping mode	Second Class	82.9%
3	Order status	SUSPECTED_FRAUD	82.1%
4	Order status	PAYMENT_REVIEW	80.0%
5	Order status	PENDING	69.1%
6	Payment type	TRANSFER	64.5%
7	Payment type	PAYMENT	63.6%
8	Order status	PENDING_PAYMENT	62.6%
9	Department	Golf	61.4%
10	Department	Outdoors	61.0%

East Africa — cross-check

Running the identical drill-down on East Africa returns the same hierarchy with different categories on top: **First Class (100.0%)**, **PAYMENT_REVIEW (83.3%)**, **Second Class (79.2%)**, **CASH / CLOSED (67.9%)**, **ON_HOLD (64.8%)**, **PENDING (61.5%)** and the **Fitness department (61.5%)**. The pattern is stable across regions: express shipping and unsettled payments are the two universal causes.

Interpretation

- **Shipping mode** dominates the top of every driver list — an SLA / routing configuration problem, not a capacity problem.

- **Payment states** (PAYMENT_REVIEW, PENDING, PENDING_PAYMENT, SUSPECTED_FRAUD) delay dispatch before goods ever leave the warehouse — a workflow and controls bottleneck.
- **Department-specific gaps** (Outdoors & Golf in Central Africa, Fitness in East Africa) point to category-level inventory or carrier handling issues worth a targeted audit.
- The same function can be re-run for **any region, segment or dimension**, making it a repeatable monthly diagnostic.

8. Time-Based Analysis

Delivery risk was profiled by month, day of week and hour of day to test whether delays are driven by calendar effects — festival seasons, weekend backlog, shift handovers — rather than by geography.

Time-Based Delivery Analysis

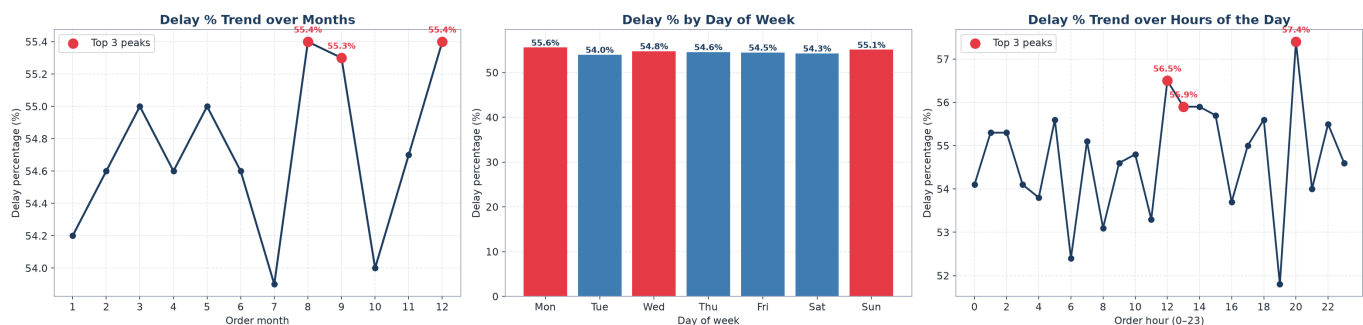


Figure 5 — Delay percentage by month, day of week and hour of day (top-3 peaks highlighted in red).

Lens	Worst periods (delay %)	Best period	Operational reading
Month	August 55.4%, December 55.4%, September 55.3%	July 53.9%	Year-end and late-summer peaks exceed planned capacity
Day of week	Monday 55.6%, Sunday 55.1%	Tuesday 54.0%	Weekend order backlog is carried into Monday dispatch
Hour of day	20:00 (8 PM) 57.4%, 12:00 56.5%, 13:00 55.9%	19:00 51.8%	Evening order wave and midday shift changeover are the stress points

Seasonality is real but modest: the spread between the best and worst month is only ~1.5 points, yet it repeats every year — August, September and December are consistently the pressure months. The hourly curve is far more volatile (51.8% → 57.4%), which means **intra-day capacity and shift planning is the higher-yield lever**.

9. Predictive Model

To move from reporting to prevention, a binary classification model predicts whether an order will be delivered late using only information available **at the moment the order is placed**: scheduled shipping days, payment type, category, customer segment, department, region, shipping mode, order month and order hour.

Method

- **Encoding:** frequency encoding for the six high-cardinality categoricals (one-hot would explode the feature space — 24 regions, 50 categories).
- **Split:** 80 / 20 stratified train-test split (random_state = 42).
- **Imbalance handling:** SMOTE oversampling applied *only* to the training set — 62,280 on-time vs 79,181 late orders became a balanced 79,181 / 79,181; the test set stays untouched.

- **Algorithm:** Random Forest (100 trees, random_state = 42), chosen after other classifiers underperformed.
- **Evaluation:** accuracy, precision, recall and a full classification report.

Metric	Result	Reading
Accuracy	73.72%	≈74% of unseen orders classified correctly
Precision (late class)	77.88%	When the model warns, it is right ~4 times in 5
Recall (late class)	74.09%	Captures roughly three quarters of actual late orders
F1 (late class)	0.76	Healthy balance of precision and recall
Test set	35,366 orders	Held out, never seen during training or balancing

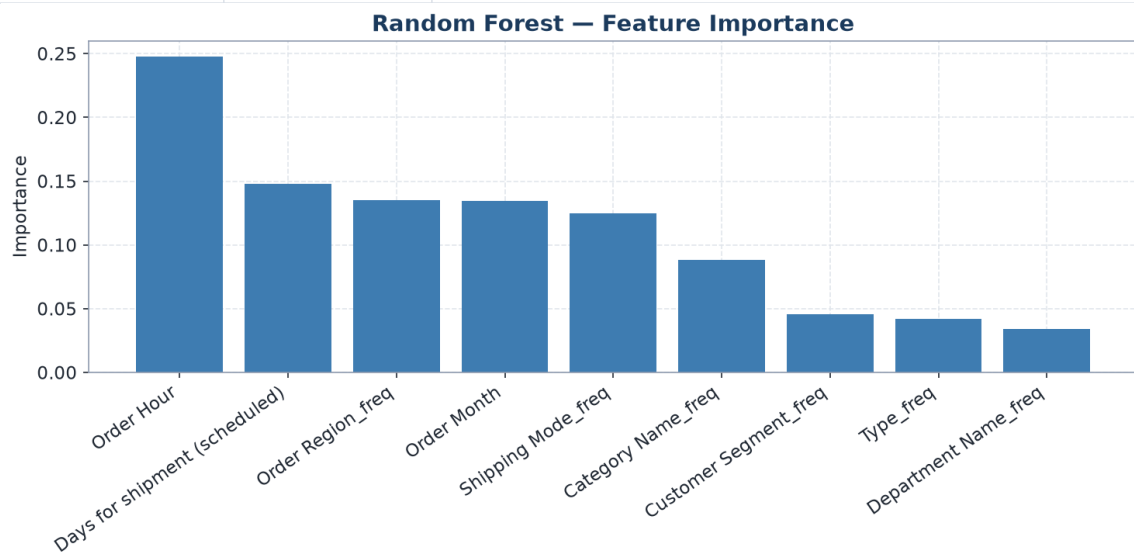


Figure 6 — Random Forest feature importance (top drivers of delay risk).

Business value: the model is not a research artefact — deployed at order capture it produces a risk score for every incoming order, letting operations re-route, pre-pick or escalate the ~30% of orders most likely to fail *before* the shipment leaves the dock. Because it trains on fields the business already stores, it can be retrained quarterly and scored in real time.

10. Strategic Recommendations

The recommendations below are ordered by expected impact and speed of implementation. Each one is anchored to a specific finding in this report.

#	Recommendation	Evidence	Expected impact
1	Audit First Class and Second Class shipping capacity and routing rules immediately	First Class 100% delay, Second Class 79.8% (Fig. 3, Fig. 4)	Removes the single largest structural failure; fastest route to a measurable on-time gain
2	Deploy the predictive alert system at order capture	Random Forest: 73.7% accuracy, 77.9% precision (§9)	Shifts operations from reactive to proactive; at-risk orders flagged before dispatch
3	Resolve payment-processing bottlenecks — auto-escalate payment reviews older than 2 hours	PAYMENT_REVIEW 80%, PENDING 69.1%, PENDING_PAYMENT 62.6% in Central Africa (§7)	Eliminates the second-largest driver; removes warehouse-side waiting time

#	Recommendation	Evidence	Expected impact
4	Develop seasonal surge capacity planning for August, September and December	Delay peaks at 55.4% in Aug/Dec vs 53.9% in July (Fig. 5)	Flattens the recurring year-end spike without permanent cost increases
5	Default eligible orders to Standard Class	Standard Class 39.7% vs First Class 100% delay (Fig. 3)	Improves promise reliability and reduces premium-freight spend
6	Investigate high-delay departments in Africa (Outdoors, Golf, Health & Beauty)	Health & Beauty 57.4%, Outdoors 61.0%, Golf 61.4% (Fig. 3, Fig. 4)	Category-level inventory / carrier fixes for the hardest-hit product lines
7	Reduce the loss-making order share (18.7%)	1 in 5 orders destroys value; \$2.1M at risk on delayed orders (\$3, \$4)	Direct margin recovery — the average order earns only \$22.01
8	Retrain the predictive model quarterly with carrier, weather and warehouse-utilisation data	Current recall 74.1% (\$9)	Target recall >80% on late orders within six months; continuously improving early warning

11. Conclusion & Current vs Target State

The analysis surfaces a clear and urgent picture of a global e-commerce operation where the majority of orders (54.71%) fail to meet their promised delivery windows, placing **\$2.1 million** of profit at risk. The root causes are identifiable and addressable: **shipping-mode misconfiguration is the dominant operational failure** (First Class ships at a 100% delay rate), **payment-processing friction creates a secondary bottleneck** (PAYMENT_REVIEW and PENDING_PAYMENT orders), **seasonal volume spikes in August, September and December are not covered by capacity planning**, and **geographic disparities are secondary** to the systematic, company-wide pattern — every region and segment sits between 53% and 59%.

A Random Forest model trained on readily available order features already achieves 74% accuracy, making the predictive layer **deployable today rather than a future aspiration**. Combined with the routing, payment and seasonal recommendations above, the business can realistically move from a majority-late operation to a majority-on-time one within two to four quarters.

Current state vs target state

Metric	Current state	Target state (6–12 months)	Primary lever
Late delivery rate	54.71%	< 35%	Routing audit + payment escalation
On-time delivery rate	45.29%	> 65%	Express-tier reconfiguration
First Class delay rate	100%	< 20%	Recalibrate SLA scheduling
Second Class delay rate	79.8%	< 40%	Carrier capacity review
Profit at risk (delayed)	–\$2.11M	–\$1.0M or better	On-time recovery + loss tail reduction
Loss-making order share	18.7%	< 12%	Pricing / cost-to-serve review
Model recall on late orders	74.1%	> 80%	Quarterly retrain + richer features
Peak-month delay (Aug/Dec)	55.4%	< 52%	Seasonal surge capacity

Next steps. (1) Socialise this report with supply-chain and finance stakeholders to agree which KPIs belong on the executive dashboard. (2) Pilot the First Class routing audit in Central and East Africa — the two highest-delay corridors. (3) Operationalise the predictive model as a daily at-risk order list. (4) Re-run this notebook monthly to track movement against the target state above.

Appendix — Methodology & artefacts

Item	Detail
Dataset	DataCo Smart Supply Chain Dataset (Kaggle), 180,519 × 53, latin-1 encoding
Pipeline	Load → inspect → clean (33 columns dropped, 3,692 cancelled removed) → feature engineering → KPIs → profitability → bottlenecks → root cause → time analysis → predictive model
Notebook	Supply_Chain_Delivery_Performance_Analysis.ipynb (fully executed, 6 exported charts)
Charts	charts/01_profitability_distribution.png ... charts/06_feature_importance.png
Reproducibility	Fixed random seeds (random_state = 42) for train/test split, SMOTE and Random Forest